Appendix A. Complete catalog, source tiers, and evidence/implementation boundaries

Appendix A. Complete catalog, source tiers, and evidence/implementation boundaries I

This appendix is the authoritative rendered snapshot of DuckRabbit's live catalog. It is generated from the taxonomy registry and evidence matrix during the publication build; the table is not hand-maintained. The snapshot records the package's current scope, not a claim to enumerate every illusion described in the literature.

How to read the matrix I

Each row identifies a registered illusion family and separates five questions that are often collapsed in informal catalogs:

1. What modality, mechanism, and signature does the package assign to the construction?
2. What evidence and implementation status does the checked-in record support?
3. Which sources support the exact statement written in the row?
4. What physical claim is supported by the row?
5. What full source role, resolver, DOI status, and claim level are retained in the machine-readable record?

How to read the matrix II

`implemented` means that DuckRabbit has a deterministic generator and a validated canonical artifact contract. It does not mean that the generated stimulus is a pixel-identical reproduction of a historical experiment or that an observer effect has been re-established. `input_required` means that the family remains catalogued but lacks a lawful, checksummed input fixture and/or the validation contract required for deterministic generation. Evidence status and implementation status are deliberately independent.

The source column is a compact citation-key view. The full source role, resolver or DOI status, exact supported claim, engineering departure, and limitation are retained in the machine-readable evidence matrix and in the generated evidence audit [@tbl:evidence_audit]. A source supports only the narrow claim recorded for it; a review or theory record is not evidence that a DuckRabbit rendering produces a universal perceptual effect.

How to read the matrix III

Table 1: DuckRabbit catalog entries, source tiers, and evidence/implementation boundaries.

ID	Construction	Evidence / status	Sources	Supported claim
visual.duck rabbit	visual; ambiguity; bistability	literature backed engineering entry; implemented	brugger 1999 duckrabbit; gregory 1997 visual	The entry instantiates a controllable ambiguous-figure stimulus family.
visual.simultaneous contrast	visual; contrast and context; contrast distortion	literature backed engineering entry; implemented	gregory 1997 visual	The physical center patches are matched while surround luminance differs.
visual.apparent motion	visual; temporal motion; illusory motion	literature backed engineering entry; implemented	wertheimer 1912 motion; sekuler 1996 wertheimer	The sequence contains controlled successive spatial events.
audio.shepard tone	auditory; spectral harmonic; continuity	literature backed engineering entry; implemented	shepard 1984 scale	The audio contains octave-related partials with controlled sweep parameters.
audio.missing fundamental	auditory; spectral harmonic; filling in	literature backed engineering entry; implemented	zatorre 2005 missing	The nominal fundamental component is absent from the canonical spectrum.
audiovisual.sound induced flash	audio_visual; crossmodal temporal; fusion or fission	review backed engineering entry; implemented	shams 2000 sifi; hirst 2020 sound	The declared beep/flash event counts and offsets are physically encoded.
audiovisual.ventriloquist	audio_visual; crossmodal spatial; spatial capture	review backed engineering entry; implemented	bruns 2019 ventriloquist; noppeney 2018 causal	The audio and visual channels carry a declared spatial discrepancy.

How to read the matrix IV

visual.muller lyer	visual; geometric alignment; geometric distortion	literature backed engineering entry; implemented	gregory 1997 visual; howe 2005 muller	The two bar lengths are equal in the canonical raster while wing geometry varies.
visual.poggendorff	visual; geometric alignment; geometric distortion	literature backed engineering entry; implemented	gregory 1997 visual; morgan 1999 poggendorff	The occluder and diagonal continuation are generated from explicit geometry.
visual.ponzo	visual; geometric alignment; geometric distortion	literature backed engineering entry; implemented	fisher 1967 ponzo; yildiz 2022 ponzo	Target bars are physically equal and rails converge toward a vanishing region.
visual.kanizsa triangle	visual; contrast and context, geometric alignment; filling in	literature backed engineering entry; implemented	kanizsa 1976 contours; wagemans 2012 gestalt	The image contains incomplete inducers with no explicitly drawn triangle edge.
visual.ebbinghaus	visual; contrast and context, geometric alignment; geometric distortion	literature backed engineering entry; implemented	mruczek 2015 ebbinghaus; weintraub 1979 ebbinghaus	Central target geometry is held equal while contextual circle geometry differs.
visual.zollner	visual; geometric alignment; geometric distortion	literature backed engineering entry; implemented	zoellner 1860 pseudoscopy; earle 1995 zollner; gregory 1997 visual	The entry instantiates a controlled crossing-line orientation stimulus family.
audio.tritone paradox	auditory; spectral harmonic; categorical recoding	literature backed engineering entry; implemented	deutsch 1986 tritone; repp 1997 tritone	The canonical pair has an explicit half-octave frequency relation.
audio.octave illusion	auditory; stream segregation; fusion or fission	literature backed engineering entry; implemented	deutsch 1974 octave	The two channels receive alternating octave-related tones.
audio.auditory continuity	auditory; stream segregation; continuity	literature backed engineering entry; implemented	warren 1970 continuity; riecke 2011 continuity	The canonical audio contains a reproducible interruption interval and masker family.

How to read the matrix V

audiovisual.temporal ventriloquism	audio_visual; crossmodal temporal; temporal binding or recalibration	review backed engineering entry; implemented	vroomen 2004 temporal; hartcherobrien 2011 temporal; hirst 2020 sound; noppeney 2018 causal	Audio and video event timing and declared offset are deterministic and inspectable.
audiovisual.mcgurk	audio_visual; speech categorization; categorical recoding	literature backed input dependent entry; input required	mcgurk 1976 speech	The catalog identifies a speech-dependent audiovisual family without claiming implementation.

Evidence boundary and future expansion I

The matrix is intentionally a living registry snapshot. New families may be added when the package can state a typed parameter contract, deterministic canonicalization rule, validation invariant, and evidence boundary. Planned or input-dependent families are not silently promoted because a source exists: promotion requires an implementable contract, reproducible fixtures where needed, and an explicit statement of what remains unvalidated. McGurk therefore remains `input_required` pending a consented or licensed, checksummed speech/ video fixture and a study-ready validation protocol. The candidate-future catalog is documented separately from this live matrix so that absence is not misread as a claim that a phenomenon is unknown or unimportant.